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Applied Statistical Methods: Including Nonparametric and Bayesian Approaches.

Inhan Kang*

Department of Psychology, Yonsei University, 50 Yonsei-ro, Seodaemun-gu, Seoul 03722, Republic of Korea

*qpsy@yonsei.ac.kr

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Statistical literacy is a core component of most undergraduate and graduate training in the social sciences. Nonetheless, the transition from coursework to independent research often reveals a persistent gap. One contributing factor is the mismatch between the assumptions emphasized in introductory courses and the properties of real-world data. Empirical data are often irregular or skewed, rather than conforming to the familiar bell-shaped form assumed by many canonical statistical procedures. When model assumptions are violated, researchers may struggle to identify appropriate alternatives.

Applied Statistical Methods: Including Nonparametric and Bayesian Approaches, by W. Holmes Finch, aims to address this pedagogical gap. By offering an integrated treatment of parametric techniques alongside corresponding alternatives within common research scenarios, the book seeks to broaden the methodological horizon at the introductory level. Although numerous texts on nonparametric and Bayesian methods are already available, they frequently emphasize technical detail that may be inaccessible to readers with limited mathematical training. From this perspective, this book represents an ambitious and potentially valuable attempt to introduce alternative approaches in a more accessible manner. In particular, it may serve as a preliminary roadmap for readers who wish to explore the available options before engaging with more mathematically demanding treatments.

The book can be broadly divided into three parts. The opening chapters introduce the scope of the book and lay out the fundamental concepts and principles that underpin the subsequent discussion (Chapters 1 and 2). The middle section focuses on methodological approaches to research questions involving population-level location (central tendency), covering a range of commonly encountered scenarios (Chapters 3-7). The final section turns to methods for analyzing associations between variables (Chapters 8-13). Within the last section, the book moves from a basic introduction to correlation and regression (Chapters 8-10) to more advanced

topics and extensions beyond linear regression (Chapters 11-13). In addition, much of the material in Chapters 3-10 follows a consistent pedagogical sequence, beginning with motivating examples and data exploration, progressing through traditional parametric methods, explicit assessment of underlying assumptions, and alternative nonparametric or Bayesian methods when those assumptions do not hold. This organizational choice encourages readers, within a unified framework, to consider a range of methodological options applicable across diverse research contexts.

The first two chapters set the stage by reviewing fundamental statistical concepts. Chapter 1 briefly revisits core topics from introductory statistics, moving from levels of measurement to null hypothesis significance testing in a concise manner. Chapter 2 then introduces the primary focus of the book on nonparametric approaches, presenting rank-based methods, trimming and Winsorizing, M-estimators, bootstrapping, and permutation techniques. The chapter also introduces Bayesian estimation as an alternative framework, illustrated through the problem of estimating a single population mean.

Following the introductory chapters, the book devotes substantial attention to research questions about location across a broad range of study designs, including the one-population case (Chapter 3), comparisons between two (Chapter 4) or more independent groups (Chapter 5), factorial designs (Chapter 6), and repeated-measures and mixed designs (Chapter 7). Throughout these chapters, parametric methods (e.g., the t-test and ANOVA) are introduced first, followed by an explicit discussion of their key assumptions, such as normality and homoscedasticity. Then, the chapters present tools for assumption checking, including Q-Q plots, the Shapiro-Wilk test, the Levene test, and related extensions for different research scenarios. Using motivating examples that violate commonly invoked assumptions, the chapters introduce appropriate nonparametric approaches, beginning with Welch's methods for heteroscedasticity and applying the nonparametric ideas introduced in Chapter 2 (e.g., rank-based methods, trimming and Winsorizing, etc.) to problems of location.

Similar nonparametric ideas recur across chapters, tailored to each study scenario, allowing readers to gradually familiarize themselves with a range of nonparametric approaches. The text further describes which methods are more appropriate under particular violations of assumptions, discusses their limitations, and outlines possible alternatives, thereby naturally exposing readers to a variety of plausible empirical situations and available methodological choices. At the same time, one potential concern is that the absence of software-based computational examples could somewhat constrain the pedagogical impact of this material. Although the book provides small-sample hand-calculation illustrations (e.g., $N = 15$ or 18), it seems unlikely that the target readers can practice related calculations without computer software, except for a few simple methods. Furthermore, the coverage of Bayesian methods remains largely confined to the simplest normal-likelihood case, even in the apparent presence of outliers and skewness, which seems limited given the overarching emphasis of the book on robust estimation.

Chapter 8 marks the beginning of the second part of the book and introduces measures of association, including Pearson's r , Spearman's ρ , Kendall's τ , and three M-estimators. Measures for non-continuous data, such as the biserial correlation, ϕ coefficient, and Cramér's V , are also covered. Chapter 9 proceeds to linear regression using ordinary least squares (OLS)

estimation, outlining the key properties and assumptions of the model, followed by visual and statistical tools for assessing violations of these assumptions. This chapter also introduces several approaches to outlier detection. Robust linear regression based on nonparametric alternatives is addressed separately in Chapter 10. Most of these approaches extend techniques introduced in Chapter 2 to the regression context, as done in the previous location part. Although the chapter covers many methods, it generally provides brief, non-technical descriptions rather than detailed formula-based exposition. This approach appears consistent with the broader aim of the book, which prioritizes introducing a diverse set of extensions without overwhelming technical detail. Perhaps this aim could be more appreciated if the text provided more detailed citations for advanced readers. Besides, the discussion of Bayesian estimation remains at a rather superficial level, neither substantially engaging with likelihood choices for robust estimation nor providing computationally accessible examples. This somewhat limits its contribution to the central theme of the book.

The remaining chapters extend the scope of regression analysis in the book. Chapter 11 is devoted to cases in which the dependent variable is dichotomous, introducing standard approaches such as logistic, probit, and complementary log-log models. In addition, beta regression is discussed as an option for proportional and probabilistic outcomes. Chapter 12 covers a range of advanced issues in regression analysis. While variable selection, regularization, and nonlinear regression constitute the core of this chapter, other topics such as mediation, moderation, and dominance analysis are also briefly described. The book concludes in Chapter 13 by considering multilevel data that violate the independence assumption in linear regression. An overview of the basic structure and assumptions of multilevel models is provided, alongside methods for outlier detection and nonparametric alternatives.

While the book provides a broad and informative overview of both classical methods and their alternatives, it may be helpful to note several aspects that some readers could find limiting. One such aspect concerns the presentation of the theoretical foundations in Chapter 2, which proceeds at a relatively brief and rapid pace. The chapter introduces multiple nonparametric ideas in a compact 17-page span and relies primarily on textual explanation, without support from illustrative figures or diagrams. Given that these ideas underpin the nonparametric methods used throughout later chapters and across a wide range of research scenarios, a more detailed exposition in Chapter 2 might further enhance accessibility for the intended audience of non-statisticians (as stated in the preface). Relatedly, some readers may wish for more replicable hands-on tutorial examples. Although the publisher's website states that the book "features a website containing datasets and computer code", it is largely left to the readers to find them. The book once directs readers to the author's personal homepage (on p. 27), but I was unfortunately unable to find any book-related materials. Providing code that allows readers to replicate the reported analyses would substantially enhance the pedagogical value of the book.

The discussion of Bayesian methods may also feel somewhat limited in scope. In particular, the text does not explore using alternative likelihood specifications to achieve robustness when standard assumptions are violated (e.g., a likelihood based on a t-distribution in the presence of outliers; pp. 458 and 478 in Kruschke, 2015). Also, computational aspects such as MCMC sampling are described mainly in prose, with no additional visual support. This approach, which continues in later chapters, may be sufficient for experienced readers seeking a condensed

overview, but it may offer less guidance for those encountering Bayesian methods for the first time. Finally, a few points in the text may benefit from further clarification. For example, the book refers to the posterior probability that the population mean is less than the null mean as a p-value (e.g., pp. 58). Also, the text describes Bayes factors as if they were defined with respect to individual hypotheses (e.g., a Bayes factor for a hypothesis H_i in Equation 3.18 and Table 3.3 on pp. 60-61). The cited approach is an approximated adjusted fractional Bayes factor (Gu, Mulder, and Hoijsink 2018), with Bayes factors for specific hypotheses defined relative to an unconstrained alternative hypothesis H_a that is not explicitly defined in the book. Although these terms may be intended to aid readers, they could lead to confusion without additional explanation.

Despite these considerations, the book overall represents a thoughtful and integrative effort to present parametric and nonparametric approaches within a unified framework. For non-statisticians, it can serve as an accessible guide to surveying a range of alternative methods without the technical demands of more specialized texts. The book may prove useful for motivated readers who are willing to supplement the material with more advanced statistical references (e.g., Wilcox 2021). In addition, instructors may also find it a helpful resource when choosing which nonparametric methods to teach for students in applied fields.

Inhan Kang

Yonsei University

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